Second Order Confirmatory Factor Analysis of The Satisfaction Level's on Massive Open Online Course (MOOC) By Generation-Z in Indonesia

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Abstract. This research explains the concept of Massive Open Online Courses (MOOC) as an innovation in technology-based education, highlighting its growing popularity globally, including in Indonesia. The purpose of the study was to analyze the factors that influence the measurement of MOOC satisfaction. This study used a quantitative approach and collected data from 220 MOOC users. Data analysis was conducted using Second Confirmatory Factor Analysis (CFA) to verify the factor structure of the instrument based on the model developed in the study. The findings of the study indicate that five indicators namely Course Content, Course Delivery, Course Support, Interaction, and Networking Opportunities showed good goodness of fit criteria, indicating a valid and reliable instrument to measure MOOC satisfaction. This study concludes that MOOC are generally considered satisfactory by users. The results highlight the importance of factors such as engagement, course delivery, and content in influencing the measurement of MOOC satisfaction. The validation of the instrument used to measure MOOC satisfaction demonstrates its usefulness for future research and assessment of MOOC effectiveness. The study provides a comprehensive analysis of MOOC factors influencing satisfaction Gen Z, offering valuable insights for educators, policymakers, and stakeholders in the field of online education.

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1 Introduction

Massive Open Online Courses (MOOC) have grown in popularity as a means of delivering educational content to a large number of students. MOOC are online courses offered by reputable educational institutions through platforms like Coursera, edX, and Udemy. The term Massive Open Online Courses (MOOC) refers to an initiative focusing on a broad community. The term "massive" refers to a large scale intended for a broad audience. "Open" signifies that these resources are freely accessible or have minimal cost. They are designed to accommodate diverse demographics, encompassing various age groups, genders, educational backgrounds, and financial situations. "Online" refers to the accessibility of courses through the internet from any location. Massive Open Online Course (MOOC) market is growing annually, demonstrating the veracity of the notion that learning is necessary. The rise of Generation Z, which accounted for 26% of the world's population at the time roughly 2 billion people followed this, and marketers need to be aware of this.

The success of MOOC depends on both students' course completion rates and their satisfaction with the learning experience. User dissatisfaction can lead to low course completion rates, ranging from 7% to 20% [1]. The purpose of this study is to investigate the factors that student satisfaction in project-based MOOC models. Previous study indicated that student satisfaction and motivation with e-learning platforms, such as MOOC, fall below expected levels. Further research is necessary to understand the factors that contribute to student satisfaction in MOOC [2]. Previous study has identified various factors that influence student satisfaction with Massive Open Online Courses (MOOC). Research indicates that several factors, such as prior online learning experience, influence students' satisfaction levels. Students with prior knowledge of the course material are typically more motivated and satisfied with MOOC compared to their peers [3]. Three factors, system quality, attitude, and course quality, significantly influence students' satisfaction in the MOOC platform

The design of MOOC platforms has a big impact on teachers, students, and other users. Six areas of indicators for MOOC design have been identified by prior research: social tools, instructional design, evaluation, user interface, and video material. The most significant category among these is learning analytics. At the university level, it is important to consider course delivery and product-focused courses when examining the factors that influence MOOC participant satisfaction [4]. Studies on user satisfaction with MOOC suggest that service quality has an impact on user satisfaction [5]. In some instances, the need for new features may arise. Implementing a dialog-based or automated interaction system in the MOOC platform allows for cost-effective 24/7 responses to learners' questions [6]. Although online learning has experienced significant growth, traditional classroom teaching with in-person interaction remains the prevailing method of teaching [7]. Without proper guidance and communication, students may lack the necessary awareness to determine whether they are progressing in the correct direction or not. New learners discover this after the course is completed based on exam results. Other studies show the quality of teaching is as important as the quality of design. Designing effective online courses is crucial for educating learners. A literature review of COVID-related research identified factors that influence learner satisfaction in MOOC.

The elements consist of technology, learner support, pedagogy, content, assessment, networking possibilities, interaction, collaboration, motivation, and usability. These elements support learning and enjoyment in MOOC. High-speed internet and smart devices have a positive impact on learners' adoption of MOOC. Interactive channels, feedback sharing, and social learning have a positive impact on an individual's perception of MOOC [8]. The primary goals of MOOC participants are influenced by personal interest and lifelong learning. Instructor interaction is crucial for learner retention in MOOC.

In Indonesia, the number of productive generations exceeds that of non-productive generations. The data shows the distribution of different generations. The pre-boomer generation (born before 1945) is 1.87%, the baby boomer generation (born from 1946-1964) is 11.56%, the Gen X generation (born from 1965-1980) is 21.88%, the Millennial generation (born from 1981-1996) is 25.87%, the Gen Z generation (born from 1997-2012) is 27.94%, and the generation born from 2013 onwards is 10.88% [9]. The data shows that Generation Z is the dominant demographic group in Indonesia. Students from different generations have unique characteristics influenced by their childhood experiences, which influences how they perceive learning. Research findings show a substantial increase in use of MOOC in Indonesia between 2017 and 2019, with a user base of 695,000 individuals. When integrated with paid platforms, the user reaches 25,453,000. This highlights the significant potential for the growth of MOOC in Indonesia, in line with Regulation Number 119 of 2014 by the Ministry of Education and Culture, that encourages open distance learning as an alternative educational. However, study on MOOC in Indonesia is limited, as previous studies have focused on technical recommendations rather than exploring their actual implementation. This study aims to provide solutions for the development of MOOC, specifically in Indonesia. The research aims to understand the factors that influence Generation Z's satisfaction with MOOC and provide recommendations for future MOOC implementation. This study presents a novel application of Second-Order Confirmatory Factor Analysis (CFA) to measure the satisfaction of Generation Z users with MOOCs in Indonesia—a population underrepresented in prior MOOC literature. Unlike previous studies that focus on general satisfaction metrics or technical implementation, this research uniquely focuses on Generation Z's perception, validating a comprehensive model of satisfaction indicators including content, delivery, interaction, support, and networking. The findings provide critical insights for optimizing MOOC design to suit the distinct learning preferences of digital-native learners in developing regions.

The literature review explains factors that influence the satisfaction of MOOC participants. This study examines participant satisfaction following the completion of MOOC, with a focus on identifying the factors that influence satisfaction levels. The significance lies in the rapid progress of MOOC development. Furthermore, MOOC remain an appropriate choice for post-COVID education and are likely to continue indefinitely. This study aims to identify factors that influence user satisfaction, such as course content, delivery, interaction, support, and networking opportunities. And which facts have the greatest influence on MOOC user satisfaction.

2 Methodology

Confirmatory factor analysis (CFA) with two orders was used in this study's quantitative research design to examine the association between a set of latent variables (factors) and several variables that were observed in the instrument. The measuring variables employed in this investigation were modified from earlier research and tailored to the particular setting. These factors include Course Content, Course Delivery, Course Support, Interaction, Network Opportunity, and Motivation. Each feature was evaluated using a four-point Likert scale: Very Low (VL), Low (L), High (H), and Very High (VH). This study used primary data as the data source. The sample size for this study includes 220 users of MOOC, with 48% male and 52% female. The age of the respondents is 18-20 years. The research respondents are located in East Java. This study used primary data as the data source. The sample size for this study includes 220 users of MOOC, with 48% male and 52% female. The age of the respondents is 18-20 years. The research respondents are located in East Java. Data analysis approaches included reliability, and second-order confirmatory factor analysis methodologies. The characteristics of the students who participated in this study as respondents were ascertained through the use of descriptive analysis. Concurrently, second order confirmatory factor analysis was employed to create the appropriate tool for assessing MOOC as suggested by the model.

The Bartlett test and the Kaiser-Meyer-Olkin measure are two assumptions that must be satisfied while doing a CFA analysis. Bartlett's sphericity test (2317.1181 at $p = 0.000$) and the Kaiser-Meyer-Olkin sampling adequacy score (KMO = 0.9392) both showed that the data had enough inter-item correlations to support factor analysis.

3 Result and Discussion

3.1 Reliability

Table 1 presents the dependability coefficients for the four scales. Based on a sample of 220 respondents, the average of all responses to the scale's questions was used to compute the scale score. It can be shown that the average participant supplied a positive word scale and scored between 3,231 and 3,330. The indicator with the highest average was Course Content. This shows that the execution of MOOC participant satisfaction is regarded as the best when compared to other metrics. Furthermore, the Cronbach's alpha coefficients for each indication were greater than 0.70, indicating that the scales had acceptable internal reliability.

Table 1. Mean, Variance of a Cronbach Alpha for MOOC Satisfaction Instruments

No	Indicators	Mean	Variance	Cronbach Alpha
1	Course Content	3,401	0,637	0,825
2	Course Delivery	3,330	0,655	0,891
3	Interaction	3,328	0,682	0,922
4	Network Opportunity	3,308	0,666	0,892
5	Course Support	3,231	0,714	0,949

3.2 Measurement Model Test

In CFA, the first step that n CFA, the first step that must be done is to test the measurement model. The tests include the convergent validity test, discriminant validity test, and reliability test. Convergent validity test is done by measuring the loading factor value in the model. Indicator measurement is declared valid if the Standardized Factor Loading (SFL) > 0.50. Invalid indicators (SLF 0.5) are included for further estimation. The calculation results are shown as follows in Table 2.

Table 2. Standardized Factor Loading (SFL)

Variable	Measurement Items	SFL	T-value	Description
Course Content	Course content Visible (CC 1)	0,717	8,604	Valid
	Course content interesting (CC 2)	0,844	8,978	Valid
	Course content simple (CC 3)	1,100	10,693	Valid
	Course content useful (CC 4)	0,963	9,466	Valid
	Course content relevant (CC 5)	0,965	9,467	Valid
	Course content structure (CC 6)	1,000	-	Valid
Course Delivery	Presenter delivered in a structured (CD 1)	0,951	10,215	Valid
	Presenter delivered it in an interesting and enjoyable (CD 2)	1,070	10,535	Valid
	Presenter delivered it was easy to understand (CD 3)	0,963	9,716	Valid
	Presenter helps complete the project (CD 4)	1,000	-	Valid
Course Support	Platform Support in completing projects (CS 1)	0,810	9,564	Valid
	Video conference platform support in course (CS 2)	1,039	13,134	Valid
	Information is easy to access (CS 3)	1,069	12,543	Valid
	Platform supports the course process (CS 4)	1,070	13,446	Valid
	Platform easy to use (CS 5)	1,000	-	Valid
Interaction	Presenter growing positive interactions (IT 1)	0,901	8,222	Valid
	Interaction in course (IT 2)	0,962	8,956	Valid
	Interaction between participants (IT 3)	0,976	12,072	Valid
	Interaction after the course (IT 4)	1,000	-	Valid
Network Opportunity	Network opportunity in course (NO 1)	1,070	9,783	Valid
	Network opportunity among participants in completing projects (NO 2)	1,009	8,937	Valid
	Network opportunities in the future (NO 3)	1,000	-	Valid

Table 2 displays the Standardized Factor Loading (SFL) values for the latent variable and indicator that are more than 0.5. This signifies that the model being generated has a valid indication. In the latent variable, the overall SLF scores vary from 0.919 to 0.997, with the greatest rating at Course Content (CD) and the lowest at Course Support (CS).

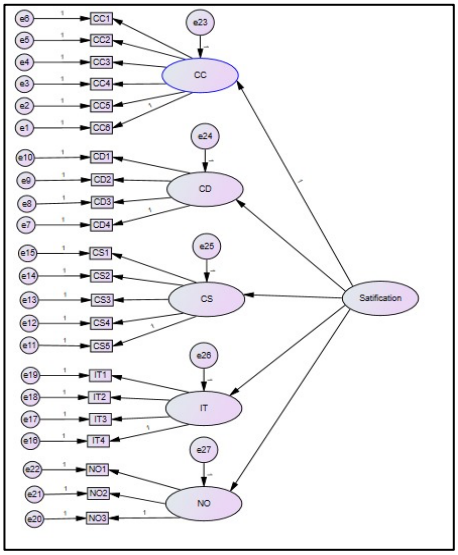

Fig. 1. Standardized Factor Loading

The discriminant validity test is then performed by comparing the square root of the AVE of each variable to the correlation value between the variables in the model. The discriminant validity test is passed if the square root value of AVE is larger than the correlation value between variables, or AVE > 0.5. For comparison, the minimal composite reliability value limit for a dependable instrument is 0.6, however 0.7 is recommended.

Table 3. Average Variance Extracted (AVE) and Composite Reliabilty (CR) of Variables

No	Indicators	AVE	CR
1	Course Content	0,713	0,925
2	Course Delivery	0,696	0,901
3	Interaction	0,679	0,894
4	Network Opportunity	0,614	0,827
5	Course Support	0,584	0,893

Table 3 shows the Average Variance Extracted (AVE) and Composite Reliability (CR) values for each hidden variable. The AVE values range from 0.584 to 0.713 and are larger than 0.5. According to the AVE value, the latent variables are discriminately valid. Meanwhile, the CR values vary between 0.827 and 0.925. These results surpass 0.7,

suggesting that the latent variable has discriminant validity according to the composite reliability value.

3.3 Confirmatory Factor Analysis

The following test result with a significance level of 0,05 (one-tailed) so that the results of the inner model evaluation are obtained.

Table 4. T Statistics for MOOC Satisfaction Instruments

No		T Statistics	P-value
1	Course Content --> satisfaction	2.628	0,052
2	Course Delivery --> satisfaction	2,123	0,009
3	Interaction --> satisfaction	3.168	0,001
4	Network Opportunity --> satisfaction	3,832	0,000
5	Course Support --> satisfaction	3,820	0,034

Based on table 4, it appears that the value of t-loading for Course Content, Course delivery, Interaction, network opportunity and course support have a significant effect on user satisfaction.

3.4 Overall Model Fit

This test was performed using the goodness of fit indicator, as indicated in Table 4. The Root Mean Square Error of Approximation (RMSEA) is a metric that may be used to account for the chi-square statistic in large samples. The RMSEA value represents the model's predicted goodness of fit when calculated in the population. An RMSEA value of less than or equal to 0.08 is required for the model to be accepted, indicating that it meets the good fit criterion and is based on degree of freedom.

Additionally, the GFI value may be used to assess the model's fit. The Goodness of Fit Index (GFI) is a non-statistical metric with values ranging from 0 (poor fit) to 1.00 (excellent fit). A high number in this indicator implies that the model is well fitted. Based on the analytical results, the GFI value was 0.71, indicating that the model in this study was in the almost excellent fit group. AGFI (Adjusted Goodness of Fit Index) values of 0.90 or above indicate the desired degree of acceptability. GFI and AGFI are criteria that consider the weighted proportion of variance in a sample covariance matrix. A rating of 0.95 might be considered an excellent level.

Table 5. Indicators of Goodness of Fit

Indicators of Goodness of Fit	Cut-off-Value	Test	Description
Root Mean Square Error of Approximation (RMSEA)	≤ 0,080	0,030	Good Fit
Normed Fit Index (NFI)	≥ 0,90	0,920	Goof Fit
Adjusted Goodness of Fit Index (AGFI)	≥ 0,90	0,838	Almost Good Fit
Goodness of Fit Index (GFI)	≥ 0,90	0,885	Almost Good Fit

The Overall Model Fit test yielded an RMSEA value of 0.095, indicating the model was virtually good fit. This is supported by the GFI and AGFI values, which are close to 0.90. This indicates that the model has a high level of acceptance and falls within the category of almost excellent fit. The model is designed to explain empirical data obtained.

The findings of this study show that Gen Z users are satisfied with learning through MOOC, which is consistent with the findings of a comprehensive review [10], of technology use in education. The study also finds that the three most important criteria determining user satisfaction with MOOC among Gen Z are course content, course delivery, and interaction. This underlines the importance of improving these three criteria in MOOC implementation to ensure that MOOC participants, particularly Generation Z, are satisfied with their learning. This finding is consistent with earlier research, which indicates that course content quality, teacher quality, and engagement involving peer-to-peer and peer-to-instructor interactions all influence MOOC acceptability [11]. The course content becomes the most significant factor capable of influencing the measurement of user satisfaction in MOOC among Generation Z.

Based on this analysis, it is proven that Generation Z MOOC users will feel satisfied if the course content is presented with a simple, attractive, well-structured, and easily understandable design, as well as being useful for personal development. The findings of this research align with [12], Involving 220 Generation Z students, the study found that Gen Z prefers learning environments that are engaging and visually appealing, integrating videos, PowerPoint slides, and other visual media. In line with this, Generation Z expresses a preference for learning methods that encourage collaboration through social media, reinforced by the use of visual media such as videos [13]. In line with study by [14] stating that providing quality content is crucial in any form of learning, especially for online learning platforms like MOOC, where instructors may not always be available to clarify misunderstandings about course content. Furthermore, other research findings show that the characteristics of teachers and the quality of instructional content are positively associated with user. Similarly, these findings suggest that service quality, including the quality of instructors and content, serves as a strong predictor for participants' perceived satisfaction with MOOC [15]. However, in addition to that, content design is identified as one aspect of content quality with a positive impact on satisfaction. These results are consistent with research conducted in the context of mobile learning, found that the quality of learning content is a primary determinant of perceived benefits and ease of use.

This also indicates that the quality of website design is one of the main factors that enhance the comfort and usability of the e-learning system. Therefore, when an e-learning system supports various types of learning content that meet the needs of students, it motivates them to use it. Another aspect related to course content is the relevance of the course content. Relevance is a major driver of learning success. Several studies have shown that learning materials that students believe to be relevant to their jobs or personal interests successfully encourage them to participate in the learning process [14]. This is because in MOOC, teachers are not always available for direct guidance in the learning process. Therefore, the main strength is course content. The results of the analysis show that the creation of course content for MOOC considers various conditions. Course content should be designed to be easily comprehensible, as

teachers have limited involvement in the learning process. In addition, it is important to ensure that the materials are well-organized. Furthermore, the material developed should be useful, particularly in facilitating the educational journey. Content relevance helps students in problem-solving during the learning process. Finally, the content should be both interesting and visually appealing or easily comprehensible.

4 Conclusion

This study examined participants' learning experiences with MOOCs and the factors influencing user satisfaction. The results showed high satisfaction levels, with "course content" being the most influential variable. Therefore, course content should be clear, relevant, engaging, accessible, and supportive of problem-solving. Interaction between instructors and participants is also important, including providing project feedback and facilitating communication among participants during and after the learning process. This study is limited to Generation Z MOOC users in East Java, which may affect the generalizability of the results. The findings offer practical insights for improving MOOC design, especially in enhancing content, interaction, and delivery. Future research should involve broader populations, consider longitudinal analysis, and explore the impact of emerging technologies like AI and gamification on user satisfaction.

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